

A Brief Review of Core Concepts in Statistical Learning

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Abstract

We review several foundational quantities that recur throughout statistical learning: information divergences, the softmax output layer, empirical risk minimization via stochastic gradient updates, mutual information, and the Gaussian likelihood. The exposition is self-contained and standalone display equations are marked in the usual way.

1 Information Divergences

The Kullback–Leibler divergence between two continuous probability distributions P and Q with densities p and q is given by

$$D_{\text{KL}}(P\|Q) = \int_{-\infty}^{\infty} p(x) \ln \frac{p(x)}{q(x)} dx. \quad (1)$$

It is non-negative and vanishes if and only if $P = Q$ almost everywhere. When comparing discrete categorical outputs, it is common to first squash logits $\mathbf{z} \in \mathbb{R}^K$ through the softmax function

$$\sigma(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad (2)$$

which yields a point on the probability simplex.

2 Empirical Risk and SGD

For binary classification the negative log-likelihood reduces to the familiar cross-entropy loss,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \{y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)\}. \quad (3)$$

Training then proceeds by stochastic gradient descent on the expected loss,

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(f_{\theta}(x), y)], \quad (4)$$

where $\eta > 0$ is a step size and $\mathcal{D}$ is the data distribution.

3 Mutual Information

For a pair of discrete random variables X and Y with joint distribution $p(x, y)$, the mutual information quantifies the reduction in uncertainty of one when the other is observed:

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \quad (5)$$

4 The Gaussian Likelihood

Finally, many continuous models assume Gaussian residuals. The density of a univariate normal with mean μ and variance σ^2 is

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right). \quad (6)$$

Combined with i.i.d. assumptions this supports both maximum-likelihood estimation and Bayesian posterior inference.